WEEK 1 - WEEK 5

Objectives:

1. Proceed with the experimental/simulation work and data collection for bidirectional memory access.

2. Able to design and conduct the experiment/simulation to capture bus turnaround and wait-state injections.

3. Choose the appropriate technique to achieve the experiment/simulation goal with proper justification regarding architectural PPA (Power, Performance, Area) evaluation.

Activities & Summary

Simulation Design & Setup

To transition from outbound write validation to full bidirectional inbound functionality, I designed a simulation environment targeting Execute-In-Place (XIP) Quad-Read operations. The experiment required configuring the Verdi testbench to rigorously monitor the timing alignment between the AHB-Lite interface and the QSPI physical pins. The primary goal was to capture the AHB wait-state injections and observe the physical SPI pin turnaround during a CPU read request, establishing definitive visual proof of the system's inbound data fetching and reconstruction capabilities.

Technique Justification & Implementation

To prove that the hardware bridge can safely manage the extreme speed differential between the high-frequency ARM processor and the slower external flash memory, I utilized a direct absolute memory read instruction in the bare-metal C code to trigger the Finite State Machine (FSM). This technique is justified because it actively forces the hardware to autonomously inject stall signals (SYS_HREADY pulled low) and manage the bidirectional pin states (qspi_data_oen). By offloading the entire protocol generation and bus turnaround logic to the hardware bridge, this implementation ensures maximum execution speed and completely eliminates software driver overhead.

Experimental Execution & Data Collection

I executed the simulation and meticulously collected golden waveform screenshots using Verdi. The experimental results successfully demonstrated the hardware pulling the HREADY signal low the exact instant the CPU initiated a read request (HWRITE = 0). The data collection verified that the FSM automatically shifted out the 0xEC (4-Byte Quad Read) command, deasserted qspi_data_oen at the precise nanosecond to switch the physical pins to input mode, captured the incoming nibbles across the four data lanes, and reconstructed the 0xDEADBEEF word accurately onto the processor's SYS_HRDATA bus before safely releasing the CPU.

Challenges & Solutions Challenge:

The highly optimized Pure XIP architecture inherently treats every AHB transaction as a high-speed memory read or write. This rigid hardware automation prevents the CPU from transmitting arbitrary or

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specific maintenance commands (such as a 0x9F JEDEC ID fetch or a Sector Erase), thereby limiting the software flexibility typically found in standard SPI controllers.

Solution: Rather than altering the RTL to include a complex, area-consuming Dual-Mode multiplexer to support a Command Mode, I evaluated this limitation as an intentional and highly beneficial architectural tradeoff. The Pure XIP approach sacrifices software command flexibility in exchange for absolute minimum latency and a significantly reduced silicon gate count (Area). I formally consulted with my project supervisor regarding this tradeoff and proposed expanding the thesis scope. The final PPA (Power, Performance, Area) evaluation will now conduct a comprehensive architectural comparison between three distinct designs: a baseline Legacy SPI with SRAM shadowing, a commercial-style Dual-Mode controller, and my proposed custom "Pure XIP" architecture optimized for peak execution speed.

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Summary of weekly activities

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FACULTY OF ELECTRICAL ENGINEERING

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